

# FACULTY OF COMPUTER SCIENCE & ARTIFICIAL INTELLIGENCE

**Subject:** Computer Science & AI Master Paper (CS-501)  
**Examination:** Internal Master Assessment | **Duration:** 60 Mins

Total Questions: 5  
Max Marks: 25.0

## OFFICIAL ANSWER KEY & GRADING RUBRIC SPECIFICATIONS

**Q1: Explain the mathematical concept of Backpropagation in Deep Neural Networks and state the chain rule formula. [5.0 Marks]**

**[2.5 Marks] Criterion 1:** Forward pass computes loss  $L$ ; backward pass calculates partial derivatives  $dL/dw$  propagating error gradient back layer by layer.

Keywords: forward pass, loss, backward pass, gradient, partial derivative

**[2.5 Marks] Criterion 2:** Chain rule application  $dL/dw = (dL/da) * (da/dz) * (dz/dw)$  and gradient descent weight update  $w = w - \eta * (dL/dw)$ .

Keywords: chain rule, gradient descent, learning rate, eta, weight update

**Q2: Analyze the average and worst-case time and space complexity of QuickSort vs MergeSort. [5.0 Marks]**

**[2.5 Marks] Criterion 1:** MergeSort: Average & Worst Time  $O(N \log N)$ , Space  $O(N)$  for merge buffer sub-arrays.

Keywords: mergesort,  $n \log n$ , space  $o(n)$ , divide and conquer

**[2.5 Marks] Criterion 2:** QuickSort: Average Time  $O(N \log N)$ , Worst Time  $O(N^2)$  for poor pivot selection, Space  $O(\log N)$  recursion stack.

Keywords: quicksort, pivot,  $o(n^2)$ , worst case, recursion stack

**Q3: Define the ACID properties in Relational Database Management Systems (RDBMS). [5.0 Marks]**

**[2.5 Marks] Criterion 1:** Atomicity & Consistency: All-or-nothing transaction execution; maintains valid state and constraints.

Keywords: atomicity, all or nothing, consistency, valid state, rollback

**[2.5 Marks] Criterion 2:** Isolation & Durability: Concurrent transactions execute independently (locking/MVCC); committed data persists in non-volatile WAL.

Keywords: isolation, durability, concurrency, locks, committed, wal

**Q4: Explain the four core Object-Oriented Programming (OOP) principles with clear definitions.** **[5.0 Marks]**

**[2.5 Marks] Criterion 1:** Encapsulation & Abstraction: Hiding internal state behind private access modifiers; exposing public interfaces.

Keywords: encapsulation, abstraction, private, interface, hiding

**[2.5 Marks] Criterion 2:** Inheritance & Polymorphism: Subclassing parent classes; method overriding/overloading allowing dynamic dispatch.

Keywords: inheritance, polymorphism, overriding, overloading, subclass

**Q5: Describe RSA Public Key Cryptography and explain how asymmetric key pairs enable secure communication.** **[5.0 Marks]**

**[2.5 Marks] Criterion 1:** Key Generation: Prime selection  $p, q$ ; modulus  $n = p \cdot q$ , totient  $\phi(n)$ ; public key  $(e, n)$ , private key  $(d, n)$ .

Keywords: rsa, prime, public key, private key, totient, modulo

**[2.5 Marks] Criterion 2:** Encryption & Decryption: Ciphertext  $c = m^e \bmod n$  and plaintext recovery  $m = c^d \bmod n$ .

Keywords: ciphertext, plaintext, encryption, decryption,  $m^e \bmod n$